

## RESEARCH INTERESTS

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- Scientific Machine Learning
- Learning-to-Optimize
- Model Predictive Control
- Cyber-Physical-Human Systems
- Robotics and Autonomous Systems
- Distributed Optimization and Multi-Agent Systems

## EDUCATION

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| <b>University of Delaware</b><br>Ph.D. in Mechanical Engineering                                 | Newark, DE USA<br>Aug. 2021–Aug. 2025   |
| <b>Northern Arizona University</b><br>M.Sc. in Informatics (with Distinction)                    | Flagstaff, AZ USA<br>Aug. 2019–May 2021 |
| <b>Hanoi University of Science and Technology</b><br>B.Sc. in Control Engineering and Automation | Hanoi, Vietnam<br>Aug. 2014–Jun. 2019   |

## RESEARCH EXPERIENCE

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| <b>University of Pennsylvania</b><br>Postdoctoral Researcher, Department of Electrical and Systems Engineering<br>– Learning-to-Optimize and Its Applications in Control                                                                              | Philadelphia, PA USA<br>Aug. 2025–Present |
| <b>Cornell University</b><br>Visiting Graduate Scholar, Systems Engineering<br>– Learning and Control for Connected and Automated Vehicles in Mixed Traffic<br>– Development of IDS’s Scaled Smart City Robotic Testbed (Version 2.0)                 | Ithaca, NY USA<br>Aug. 2023–Aug. 2025     |
| <b>University of Delaware</b><br>Graduate Research Assistant, Department of Mechanical Engineering<br>– Learning and Control for Connected and Automated Vehicles in Mixed Traffic                                                                    | Newark, DE USA<br>Aug. 2021–May 2023      |
| <b>Northern Arizona University</b><br>Graduate Research Assistant, School of Informatics, Computing, and Cyber Systems<br>– Learning-Based Model Predictive Control with Gaussian Processes<br>– Adaptive Sampling for Mobile Robotic Sensor Networks | Flagstaff, AZ USA<br>Aug. 2019–May 2021   |
| <b>Vietnam Maritime University</b><br>Undergraduate Research Intern, School of Mechanical Engineering<br>– Research, Design, and Manufacture of a Floating Crane Testbed                                                                              | Haiphong, Vietnam<br>Sep. 2017–Mar. 2019  |

## TEACHING EXPERIENCE

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| <b>University of Delaware</b><br>Graduate Teaching Assistant, Department of Mechanical Engineering<br>– <b>Vibration and Control (MEEG 312)</b> , Fall 2021 – Laboratory Instructor<br>– <b>Dynamics (MEEG 211)</b> , Spring 2022 – Recitation Instructor | Newark, DE USA<br>Aug. 2021–May 2022 |
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## INDUSTRY RESEARCH EXPERIENCE

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**Honda Research Institute USA**

Graduate Student Intern

– Social Navigation for Multiple Robots in Crowded Environments

Ann Arbor, MI USA

Jun. 2023–Aug. 2023

## PUBLICATIONS

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### JOURNAL ARTICLES

- [1] **V.-A. Le** and A. A. Malikopoulos, “Distributed Mixed-Integer Quadratic Programming for Mixed-Traffic Intersection Control,” *Automatica*, (accepted, arXiv preprint arXiv:2504.04618).
- [2] **V.-A. Le** and A. A. Malikopoulos, “Controller Adaptation via Learning Solutions of Contextual Bayesian Optimization,” *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8308–8315, 2025.
- [3] **V.-A. Le**, B. Chalaki, F. N. Tzortzoglou, and A. A. Malikopoulos, “Stochastic Time-Optimal Trajectory Planning for Connected and Automated Vehicles in Mixed-Traffic Merging Scenarios,” *IEEE Transactions on Control Systems Technology*, vol. 33, no. 4, pp. 1403–1417, 2025.
- [4] **V.-A. Le** and A. A. Malikopoulos, “Distributed Optimization for Traffic Light Control and Connected Automated Vehicle Coordination in Mixed-Traffic Intersections,” *IEEE Control Systems Letters*, vol. 8, pp. 2721–2726, 2024.
- [5] A. Mokhtarian, J. Xu, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, **V.-A. Le**, S. Ulhas, J. Betz, S. Wilson, *et al.*, “A survey on small-scale testbeds for connected and automated vehicles and robot swarms: A guide for creating a new testbed [Survey],” *IEEE Robotics & Automation Magazine*, vol. 32, no. 3, pp. 146–163, 2024.
- [6] A. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, “A safety-prioritized receding horizon control framework for platoon formation in a mixed traffic environment,” *Automatica*, vol. 155, p. 111 115, 2023.
- [7] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “Multistep Predictions for Adaptive Sampling in Mobile Robotic Sensor Networks Using Proximal ADMM,” *IEEE Access*, vol. 10, pp. 64 850–64 861, 2022.
- [8] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “ADMM-Based Adaptive Sampling Strategy for Nonholonomic Mobile Robotic Sensor Networks,” *IEEE Sensors Journal*, vol. 21, no. 13, pp. 15 369–15 378, 2021.
- [9] **V.-A. Le**, X. H. Le, L. Nguyen, and X. M. Phan, “An efficient adaptive hierarchical sliding mode control strategy using neural networks for 3D overhead cranes,” *International Journal of Automation and Computing*, vol. 16, no. 5, pp. 614–627, 2019.
- [10] H. X. Le, T. V. Nguyen, **V.-A. Le**, T. A. Phan, N. H. Nguyen, and M. X. Phan, “Adaptive hierarchical sliding mode control using neural network for uncertain 2D overhead crane,” *International Journal of Dynamics and Control*, vol. 7, pp. 996–1004, 2019.
- [11] A. T. Le, M. C. Hoang, V. T. Pham, C. N. Luong, D. T. Vu, and **V.-A. Le**, “Adaptive neural network sliding mode control of shipboard container cranes considering actuator backlash,” *Mechanical Systems and Signal Processing*, vol. 112, pp. 233–250, 2018.

### CONFERENCE PAPERS

- [1] **V.-A. Le**, R. Tumu, and R. Mangharam, “Toward Single-Step MPPI via Differentiable Predictive Control,” in *2026 IEEE 65th Conference on Decision and Control (CDC)*, (accepted, arXiv preprint arXiv:2604.01539).
- [2] **V.-A. Le** and A. A. Malikopoulos, “Accelerating Time-Optimal Trajectory Planning for Connected and Automated Vehicles with Graph Neural Networks,” in *23rd IFAC World Congress*, (accepted, arXiv preprint arXiv:2511.20383).
- [3] **V.-A. Le**, M. Xie, and R. Mangharam, “A Hybrid Learning-to-Optimize Framework for Mixed-Integer Quadratic Programming,” in *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 331, PMLR, 2026, pp. 1055–1069.
- [4] A. Amine, K. Puri, **V.-A. Le**, and R. Mangharam, “Nonplanar Model Predictive Control for Autonomous Vehicles with Recursive Sparse Gaussian Process Dynamics,” in *2026 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2026, pp. 2088–2093.
- [5] **V.-A. Le**, P. Kounatidis, and A. A. Malikopoulos, “Combining Graph Attention Networks and Distributed Optimization for Multi-Robot Mixed-Integer Convex Programming,” in *2025 IEEE 64th Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 4440–4445.

- [6] **V.-A. Le**, B. Chalaki, V. Tadiparthi, H. N. Mahjoub, J. D'Sa, and E. Moradi-Pari, "Social Navigation in Crowded Environments with Model Predictive Control and Deep Learning-Based Human Trajectory Prediction," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 4793–4799.
- [7] **V.-A. Le**, V. Tadiparthi, B. Chalaki, H. N. Mahjoub, J. D'sa, E. Moradi-Pari, and A. A. Malikopoulos, "Multi-Robot Cooperative Navigation in Crowds: A Game-Theoretic Learning-Based Model Predictive Control Approach," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 4834–4840.
- [8] **V.-A. Le**, H. M. Wang, G. Orosz, and A. A. Malikopoulos, "Coordination for Connected Automated Vehicles at Merging Roadways in Mixed Traffic Environment," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 4150–4155.
- [9] N. Venkatesh, **V.-A. Le**, A. Dave, and A. A. Malikopoulos, "Connected and Automated Vehicles in Mixed-Traffic: Learning Human Driver Behavior for Effective On-Ramp Merging," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 92–97.
- [10] **V.-A. Le** and A. A. Malikopoulos, "Optimal Weight Adaptation of Model Predictive Control for Connected and Automated Vehicles in Mixed Traffic with Bayesian Optimization," in *2023 American Control Conference (ACC)*, 2023, pp. 1183–1188.
- [11] **V.-A. Le** and A. A. Malikopoulos, "A Cooperative Optimal Control Framework for Connected and Automated Vehicles in Mixed Traffic Using Social Value Orientation," in *2022 61st IEEE Conference on Decision and Control (CDC)*, 2022, pp. 6272–6277.
- [12] A. M. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, "Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection," in *10th IFAC International Symposium on Advances in Automotive Control*, IFAC, 2022, pp. 51–56.
- [13] **V.-A. Le** and T. X. Nghiem, "Distributed Experiment Design and Control for Multi-agent Systems with Gaussian Processes," in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 2226–2231.
- [14] **V.-A. Le** and T. X. Nghiem, "A Receding Horizon Approach for Simultaneous Active Learning and Control using Gaussian Processes," in *2021 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2021, pp. 453–458.
- [15] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, "An Efficient Adaptive Sampling Approach for Mobile Robotic Sensor Networks using Proximal ADMM," in *2021 American Control Conference (ACC)*, IEEE, 2021, pp. 1101–1106.
- [16] **V.-A. Le** and T. X. Nghiem, "Gaussian Process Based Distributed Model Predictive Control for Multi-agent Systems using Sequential Convex Programming and ADMM," in *2020 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2020, pp. 31–36.
- [17] T. X. Nghiem, T.-D. Nguyen, and **V.-A. Le**, "Fast Gaussian Process based Model Predictive Control with Uncertainty Propagation," in *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, IEEE, 2019, pp. 1052–1059.
- [18] **V.-A. Le**, X. H. Le, D. T. Vu, V. T. Pham, A. T. Le, and M. C. Hoang, "Designing an adaptive controller for 3D overhead cranes using hierarchical sliding mode and neural network," in *2018 International Conference on System Science and Engineering (ICSSE)*, IEEE, 2018, pp. 1–6.

## PATENTS

- [1] **V.-A. Le**, V. Tadiparthi, B. Chalaki, H. N. Mahjoub, J. J. D'Sa, and E. Moradi-Pari, "Systems and Methods for Robot Navigation," U.S. Patent Application Publication US 2025/0117025 A1, U.S. Patent Application No. 18/907,188, Apr. 10, 2025.

## HONORS AND AWARDS

- Jun. 2026: **IEEE ITSS Best Dissertation Award (2nd Place)**, IEEE Intelligent Transportation Systems Society
- Feb. 2026: **Penn AI Fellowship**, University of Pennsylvania
- Jul. 2025: **Best Paper Finalist (2nd Place)**, ASME Automotive and Transportation Systems Technical Committee (ATS-TC), 2025 American Control Conference
- Aug. 2023: Student Travel Award, IEEE Control Systems Society, 2023 IEEE Conference on Decision and Control

- Oct. 2022: IEEEExtreme Programming Competition, Global Rank: 949/6373; U.S. Rank: 12/99
- Apr. 2022: Student Travel Awards, University of Delaware Graduate College and IEEE Control Systems Society, 2022 American Control Conference
- Sep. 2021: Student Travel Award, IEEE Control Systems Society, 2021 IEEE Conference on Decision and Control
- Jun. 2021: Student Travel Award, IEEE Control Systems Society, 2021 IEEE Conference on Control Technology and Applications
- Aug. 2020: Student Travel Award, IEEE Control Systems Society, 2020 IEEE Conference on Control Technology and Applications
- Aug. 2019: Presidential Fellowship, Northern Arizona University
- Aug. 2018: Odon Vallet Scholarship for Undergraduate Students
- Jun. 2018: Conference Travel Award, Vietnam National Foundation for Science and Technology Development (NAFOS-TED), 2018 IEEE International Conference on System Science and Engineering
- Apr. 2015: Gold Medal, Vietnam National Mathematical Olympiad for Undergraduate Students

## STUDENT MENTORING

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- Mu Xie (MS Robotics, University of Pennsylvania) – Learning-to-optimize for mixed-integer optimization
- Kabir Puri (MS Robotics, University of Pennsylvania) – Learning-based control for autonomous driving on nonplanar terrains
- Yuntong Li (MS Electrical Engineering, University of Pennsylvania) – Mixed-integer model predictive control for bipedal locomotion
- Shamanth Sreekanth (MS Electrical Engineering, University of Pennsylvania) – Physics-informed modeling and optimization for process control
- Nandagopal Vidhu (MS Robotics, University of Pennsylvania) – Integration of differentiable robotic simulators into learning-to-optimize frameworks
- Owen Sorber (MS Computer Science, Cornell University) – Development of a containerized Docker-based software stack for advanced driver-assistance systems on autonomous robotic vehicles
- Taylor Stephens (MS Systems Engineering, Cornell University) – Deployment of the Crazyswarm multi-drone platform and development of aerial-ground autonomy demonstrations using Crazyswarm and IDS3C
- Qiming Yang (MS Systems Engineering, Cornell University) – Design and implementation of a digital simulation environment for the IDS3C testbed
- Evans McCleary (BS Mechanical Engineering, University of Delaware) – Development and maintenance of hardware and software infrastructure for robotic vehicles in the IDS3C testbed

## INVITED TALKS AND TUTORIALS

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1. **Tutorial Co-Organizer and Speaker**, “Learning-to-Optimize for Control in Robotics,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), October 2026.
2. **Lightning Talk**, “Learning-to-Optimize and Its Applications in Model Predictive Control,” 3rd Northeast Systems and Control Workshop, May 2026.
3. **Invited Seminar**, “Towards Real-Time Cooperative Trajectory Planning for Connected and Automated Vehicles in Mixed Traffic,” RGSO-CARS Seminar, University of Delaware, May 2025.

## PROFESSIONAL SERVICE

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- Professional Memberships
  - Member, Institute of Electrical and Electronics Engineers (IEEE) 2020–Present
  - Member, IEEE Control Systems Society 2020–Present

- Member, IEEE Intelligent Transportation Systems Society 2022–Present
- Member, IEEE Robotics and Automation Society 2022–Present
- Technical Committee Memberships
  - Member, IEEE Control Systems Society Technical Committee on Smart Cities 2022–Present
  - Member, IEEE Control Systems Society Technical Committee on Automotive Controls 2024–Present
  - Member, IEEE Robotics and Automation Society Technical Committee on Optimization for Robotics 2026–Present
  - Member, IEEE Robotics and Automation Society Technical Committee on Robot Control 2026–Present
- Peer Review
  - **Journals:** Automatica; IEEE Transactions on Automatic Control; IEEE Transactions on Robotics; IEEE/ASME Transactions on Mechatronics; IEEE Transactions on Intelligent Transportation Systems; IEEE Transactions on Control Systems Technology; IEEE Transactions on Vehicular Technology; IEEE Transactions on Intelligent Vehicles; IEEE Control Systems Letters; IEEE Robotics and Automation Letters; IEEE Open Journal of Control Systems; Transportation Research Part C: Emerging Technologies; Journal of the Franklin Institute.
  - **Conferences:** IEEE Conference on Decision and Control (CDC); American Control Conference (ACC); European Control Conference (ECC); IEEE International Conference on Robotics and Automation (ICRA); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE International Conference on Intelligent Transportation Systems (ITSC); IEEE Intelligent Vehicles Symposium (IV); Modeling, Estimation and Control Conference (MECC); IFAC World Congress.